

Dhia Eddine BARHOUMI

SOFTWARE ENGINEERING STUDENT @ ENIT

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EXPERTISE

- Machine Learning
- Computer Vision
- Network Automation
- Cloud & DevOps
- Software Architecture

TECHNICAL STACK

- Python (Core)
- Java / Spring Boot
- TypeScript / Angular
- Docker / Containerlab
- Terraform (IaC)
- YOLO / OpenCV
- Ansible / Jinja2
- Git / CI/CD
- SQL / NoSQL

EDUCATION

ENIT

B.Eng in Software Engineering
2023 — 2026

IPEIM

Pre-Eng: Physics & Tech
High Honors (Très Bien)
2021 — 2023

CERTIFICATIONS

NVIDIA LLM Customization
NVIDIA

CCNA Introduction
Cisco

PROFESSIONAL PROFILE

2nd-year Software Engineering student at ENIT, focused on the synergy between **AI** and **Infrastructure Automation**. Passionat about Computer Vision (YOLO), Cloud Native technologies, and building scalable systems through DevOps best practices.

WORK EXPERIENCE

Full-Stack Developer Intern

Summer 2025

Tunisie Télécom — District of Kef

- Engineered a web-based IT Asset Management platform for hardware and licenses.
- Built robust backend with **Spring Boot 3.5** & JPA; migrated from H2 to PostgreSQL.
- Crafted a responsive UI using **Angular 20** & Material Design with RBAC security.

Spring Boot Angular 20 Java 21 PostgreSQL Docker

KEY PROJECTS

Vision-to-Infrastructure Pipeline (PFA 2)

ENIT · 2025

End-to-end AI system translating network drawings into Terraform configurations.

- Optimized **YOLOv8** for 98% accuracy in detecting network nodes & connections.
- Developed a graph-extraction engine to generate structured JSON/YAML topology data.
- Automated deployments via **Terraform** and **Containerlab** for instant lab validation.

Python YOLOv8 Terraform Containerlab OpenCV

Network Automation & Monitoring

GitHub · 2025

Automating lifecycle management for virtualized network infrastructures.

- Created Jinja2-based automation scripts for massive configuration deployments.
- Implemented post-deployment monitoring to ensure network state compliance.

Python Jinja2 Ansible Network Automation

SocialMedia Microservices Architecture

GitHub · 2025

Distributed system using 7 independent services and MongoDB Atlas.

- Managed inter-service communication via a centralized **API Gateway**.
- Focused on domain-driven design and horizontal scalability.

Node.js TypeScript Microservices MongoDB

RendezvousJEE — Clinic Management

GitHub · 2025

- Built a Jakarta EE 10 application using EJB Stateless beans and JPA modeling.
- Deployed on WildFly server with Maven lifecycle management.

Jakarta EE 10 Hibernate EJB WildFly